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import streamlit as st

st.set_page_config(page_title="Black Hole
Quiz", page_icon="🕳️")

# session_state remembers values between
clicks (Streamlit reruns the
# whole script every time you click a
button, so we need this to
# remember what stage/name/score we're at)
if 'stage' not in st.session_state:
    st.session_state.stage = 'intro'
if 'name' not in st.session_state:
    st.session_state.name = ''
if 'signal' not in st.session_state:
    st.session_state.signal = 87

st.title("🕳️ WELCOME TO THE BLACK HOLE
QUIZ!")

# ----- INTRO -----
if st.session_state.stage == 'intro':
    name_input = st.text_input("What is
your name?")
    if st.button("Start Quiz") and
name_input:
        st.session_state.name = name_input
        st.session_state.stage = 'q1'
        st.rerun()

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# ----- QUESTION 1 -----
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elif st.session_state.stage == 'q1':
    st.write(f"Hello
{st.session_state.name}! Let's see how much
you know about black holes.")
    st.subheader("1. Can light escape a
black hole?")
    ans = st.radio("Your answer:", ["A)
Yes", "B) No"], index=None)
    if st.button("Submit Answer",
key="q1submit"):
        if ans == "B) No":
            st.success("Correct, good
job!")
        elif ans == "A) Yes":
            st.error("Incorrect, you are
almost there!")
        if ans is not None:
            st.info("Once light crosses the
event horizon, it cannot escape. "
                    "Gravity there is so
extreme that all paths for light bend "
                    "inward toward the
center, making escape impossible.")
            if st.button("Next Question
➡", key="q1next"):
                st.session_state.stage =
'q2'
                st.rerun()

# ----- QUESTION 2 -----
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elif st.session_state.stage == 'q2':

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        st.subheader("2. A body weighs 2.56 kg.
What is its energy ( $\times 10^8$  J)?")
        st.write("Options:", [23.04, 32.05,
23.05, 22.05, 11.05])
        ans = st.text_input("Your answer:")
        if st.button("Submit Answer",
key="q2submit"):
            if ans.strip() == "23.04":
                st.success("Wow correct, we
have Einstein here!!")
                if st.button("Next Question
➡", key="q2next"):
                    st.session_state.stage =
'q3'
                    st.rerun()
            else:
                st.error("Incorrect. Hint: use
Einstein's formula  $E = mc^2$ ")

# ----- QUESTION 3 -----
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elif st.session_state.stage == 'q3':
    st.subheader("MISSION 3: CORE BREACH")
    st.write("An unmanned probe has reached
the centre of a black hole. "
"Mission Control needs YOUR
help.")
    st.write("3. Which part of a black hole
lies at its very centre?")
    ans = st.radio("Your answer:",
["A) Event Horizon",
"B) Singularity", "C) Accretion Disk", "D)
Photon Sphere"],
index=None)

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        locked = st.checkbox("Lock my answer")
        if locked and ans is not None:
            if ans == "B) Singularity":
                st.success("Correct! The
singularity lies at the very centre of a
black hole.")
            else:
                st.error("Not quite – the
singularity is the point at the very
centre.")
            if st.button("Next Question ➡",
key="q3next"):
                st.session_state.stage = 'q4'
                st.rerun()

# ----- QUESTION 4 -----
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elif st.session_state.stage == 'q4':
    st.subheader("MISSION 4: SIGNAL
INTEGRITY")
    st.progress(st.session_state.signal /
100)
    st.write(f"Signal integrity:
{st.session_state.signal}%")
    st.write("4. As the probe nears the
event horizon, how does ITS clock look "
            "to YOU, watching safely from
Earth?")
    ans = st.radio("Transmit your answer:",
                    ["A) Speeding up", "B)
Slowing down, almost freezing",
                    "C) Unchanged", "D)
Running backward"], index=None)
    if st.button("Transmit",

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key="q4submit"):
    if ans == "B) Slowing down, almost
freezing":
        st.success("SIGNAL STABILIZED.
Correct! This is gravitational time
dilation – "
                    "the deeper
something falls into a black hole's
gravity, the slower "
                    "its time appears
to run to a distant observer.")
        st.session_state.stage = 'done'
        st.rerun()
    elif ans is not None:
        st.session_state.signal -= 20
        if st.session_state.signal <=
0:
            st.error("SIGNAL LOST. The
answer was B – gravitational time
dilation.")
            st.session_state.stage =
'done'
            st.rerun()
        else:
            st.warning(f"WRONG SIGNAL.
Integrity dropping... now at
{st.session_state.signal}%")

# ----- DONE -----
elif st.session_state.stage == 'done':
    st.balloons()
    st.header(f"Thanks for playing,
{st.session_state.name}! 🚀")
    if st.button("Play Again"):

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st.session_state.stage = 'intro'  
st.session_state.signal = 87  
st.rerun()
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